

5.3 Probability

Probability and Statistics 1 · 2026 examination · Concept guide and original exam-style practice

Understand the concept

Conditional probability $P(A \text{ given } B) = P(A \cap B) / P(B)$. Independent events satisfy $P(A \cap B) = P(A)P(B)$, which differs from mutually exclusive nonempty events. Tree branches must use updated probabilities after sampling without replacement; check totals sum to one.

Worked example: follow the reasoning

For two reds among five, probability of two reds without replacement is $(2/5)(1/4) = 1/10$.

Structured practice

Answer in the space provided. Show working and units where relevant.

- 1 A bag has three blue and two green counters; draw two without replacement. Find probability of one of each. [3]

Final answer: _____

Answers and explanation

Question 1 [3 marks]

Blue-green: $(3/5)(2/4)=3/10$ [1]; green-blue: $(2/5)(3/4)=3/10$ [1]; total $3/5$ [1].

Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or calculation error; then retry without looking at the solution.

Syllabus: Mathematics 9709, 2026–2027. <https://www.cambridgeinternational.org/Images/697427-2026-2027-syllabus.pdf>

Original FagusED teaching material. Question layout and mark presentation informed by the May/June 2026 papers supplied by the user. The practice question is newly written and is not a Cambridge past-paper question.